Prototype-Driven Structure Synergy Network for Remote Sensing Images Segmentation

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***Abstract*—In the semantic segmentation of remote sensing images, acquiring complete ground objects is critical for achieving precise analysis. However, this task is severely hindered by two major challenges: high intra-class variance and high inter-class similarity. Traditional methods often yield incomplete segmentation results due to their inability to effectively unify class representations and distinguish between similar features. Even emerging class-guided approaches are limited by coarse class prototype representations and a neglect of target structural information. Therefore, this article proposes a prototype-driven structure synergy network (PDSSNet). The design of this network is based on a core concept: a complete ground object is jointly defined by its invariant class semantics and its variant spatial structure. To implement this, we have designed three key modules. First, the adaptive prototype extraction module (APEM) ensures semantic accuracy from the source by encoding the ground truth to extract comprehensive class prototypes. Subsequently, the designed semantic-structure coordination module (SSCM) follows a hierarchical "semantics-first, structure-second" principle. This involves first establishing a global semantic cognition, then leveraging structural information to constrain and refine the semantic representation, thereby ensuring the integrity of class information. Finally, the channel similarity adjustment module (CSAM) employs a dynamic step-size adjustment mechanism to focus on discriminative features between classes. Extensive experiments demonstrate that PDSSNet outperforms state-of-the-art methods. The source code is available at https://github.com/wangjunyi-1/PDSSNet**

***Index Terms*—Mamba, prototype drive, semantic segmentation, state-space model (SSM).**

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I. Introduction

With the advancement of aerospace sensors, remote sensing images now enable fine-grained observation of the Earth's surface. Semantic segmentation, a core technology in this field, transforms pixels into structured geographical objects. Its practical applications, such as land cover mapping [1], [2], change detection [3], [4], [5], and environmental protection [6], [7], [8], [9], depend on the complete parsing of class instances.

However, existing methods exhibit limitations in the completeness of their segmentation results (as illustrated in Fig. 1). We find that this problem primarily stems from two inherent characteristics of remote sensing images. On the one hand, high intra-class variance stems from the vast differences in scale, morphology, material, and lighting among objects of the same class. This resulting feature diversity directly hinders the model from learning a unified class representation capable of encompassing all variations and is a primary cause of incomplete object segmentation. On the other hand, high inter-class similarity means that different classes of ground objects exhibit similar spectral and texture features due to their comparable material compositions. This feature ambiguity makes it easy for the model to confuse different classes, thereby resulting in erroneous segmentation.

To address the challenges posed by the aforementioned characteristics, we first analyze the current mainstream technical approaches. Models based on convolutional neural networks [10], [11], [12], [13], [14], [15], [16] and transformers [17], [18], [19], [20] have achieved great success through semantic label-driven pixel classification. However, the fixed weights of CNNs make it difficult for them to dynamically adapt to the large intra-class variance in remote sensing images. Although Transformers can model global context, their indiscriminate attention mechanism is particularly sensitive to highly similar inter-class features and, lacking explicit intra-class constraints, is prone to causing class confusion [21].

These inherent limitations have prompted researchers to recognize that class constraints must be shifted from the end-stage classification task to the upfront feature extraction process. Driven by this idea, class-guided methods have emerged, with their core strategy being the modeling of class semantics to guide feature extraction, thereby compelling the model to focus on features relevant to class semantics [22], [23], [24], [25].

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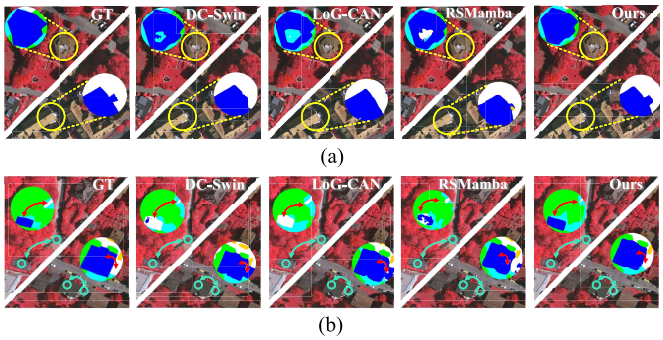


Fig. 1. This figure illustrates the problem of incomplete class segmentation, where objects that should be continuous are erroneously segmented, resulting in internal holes, fragmented overall structures, and the misidentification of one class as a similar one. We find that the root cause of this incomplete segmentation can be attributed to two main challenges. The first, as shown by the yellow circle in (a), is high intra-class variance, where significant appearance differences within the same object class make it difficult for the model to learn unified features. The second, in the region indicated by the blue arrow in (b), is high inter-class similarity, where the similar features of different classes make them easy for the model to confuse.

Although these methods have achieved significant success, they still fail to fully consider the unique characteristics of remote sensing imagery. First, they rely on coarse-grained class prototypes in the initial stage to guide the extraction of global contextual features. These initial coarse class prototypes can hardly represent class features comprehensively and introduce misleading characteristics. Second, they typically depend on class weights and self-attention mechanisms to establish the association between class information and features [26], [27]. Strategies based on class weights lose spatial details during information compression and are unable to establish contextual correlations. Moreover, the self-attention mechanisms subsequently used to establish these contextual links perform poorly in both precision and efficiency when distinguishing between similar features, owing to their inherent smoothing effect and quadratic computational complexity [28], [29], [30], [31].

The recently proposed Mamba model [32] efficiently captures long-range dependencies with linear complexity, offering an alternative to the computationally expensive self-attention mechanism. Researchers have found that by designing the order in which features enter the state-space model (SSM), the model can be guided to focus on different types of information [33]. The exploration in some recent Mamba variants has primarily focused on enhancing spatial or modal feature interaction by varying the scan order [34], [35], [36]. However, these methods generally lack a crucial mechanism for translating Mamba's sequential modeling capabilities into a refined, class-information-driven feature selection strategy. More importantly, existing class-guided methods prioritize the consistency of class semantics while neglecting structural information.

In this article, we introduce the prototype-driven structure-synergized network, which consists of a feature encoder, an adaptive prototype extraction module (APEM), a semantic-structure coordination module (SSCM), and a channel similarity adjustment module (CSAM). Traditional class modeling methods rely on the segmentation head (a 1×1 convolution) to generate coarse-grained class prototypes [23], [37], [38]. However, these coarse-grained prototypes, derived from the model's intermediate predictions, possess limited representational capability when faced with the large intra-class variance in remote sensing images. In contrast, APEM generates binary masks for different classes by encoding the GT. It utilizes the "1" values in the corresponding class masks to aggregate features of the same class across multiple channels, while simultaneously using the "0" values to suppress the influence of features from other classes. Through this approach, we are able to construct class prototypes that are both comprehensively representative and undisturbed by background interference.

Upon obtaining precise class prototypes, how to leverage them as effective prior guidance becomes another critical task. We think that for a single ground object class, its class semantics are unified and invariant, whereas its spatial structure is concrete and variable. However, the design of existing class-guided methods generally tends to reinforce semantic consistency while relatively neglecting the constraint on target structural integrity. This lack of constraint on structural integrity inevitably compromises the completeness of the segmentation. To this end, we propose the SSCM to enhance the integrity of class information. This module simulates the hierarchical cognitive strategy of humans: it first embeds class-level semantic information to enable the model to comprehend global semantics, and then progressively refines class details based on extracted structural information, allowing the model to dynamically adjust class representations according to structural differences. To effectively address the challenge of high inter-class similarity, we have designed the CSAM. This module, based on a core SimStep mechanism, enables the model to bypass local similarities by controlling its step size when encountering similar features, focusing instead on discriminative characteristics. Experimental results validate the effectiveness of the proposed method. In summary, the main contributions of this study are as follows.

1) To overcome the high intra-class variance in remote sensing images, we propose a new class modeling method, the APEM. It aggregates diverse features under a single class by leveraging the ground truth to construct a comprehensive class prototype.
2) We design the SSCM, the core of which is to drive the dynamic alignment of high-level semantic understanding with spatial structure layout. It utilizes the continuity of structural information to constrain the scope of semantic prediction, ensuring that the semantic guidance can adaptively conform to the complete contours of a class in its various morphologies.
3) We propose the prototype-driven structure synergy network (PDSSNet), a network that refines the semantic and structural representations of classes by integrating APEM and SSCM to ensure their morphological integrity. Building on this, we have designed the CSAM to enhance the final discriminative capability for highly similar classes, thereby comprehen-

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sively addressing the problem of incomplete class segmentation.

II. Related Work

A. Class-Guided Semantic Segmentation of Remote Sensing Images

Deep learning-based remote sensing image processing has made significant advancements across multiple frontiers [39], [40], [41], [42], such as effectively synergizing multisource data to enhance cross-scene generalization [43], and developing highly effective integrated frameworks to jointly optimize coupled tasks like denoising and anomaly detection [44]. In the task of semantic segmentation, mainstream technical approaches can be broadly categorized into two types: pixel-level and object-level. However, these methods face challenges in learning class representations that are both unified and highly discriminative [17], [45], [46], [47]. To address this challenge, researchers have proposed class-guidance techniques. The core idea is to provide clearer guidance for the feature extraction and optimization processes by explicitly modeling class semantic information within the network, thereby enhancing segmentation accuracy.

Most existing class-guidance methods follow a common paradigm: they utilize coarse-grained segmentation maps, generated from the network's intermediate layers, as dynamic class priors. For instance, CGGLNet [23] generates coarse segmentation results at different network stages to guide the extraction of global contextual information. CGGCNet [21] takes this a step further by using an adjacency matrix, calculated from the coarse segmentation matrix, to enforce tight connections between pixels of the same class and explicitly model contextual relationships. Another effective technical path enhances accuracy by modeling object-level contextual relationships; for example, MLCG-OCNN [48] effectively integrates multilevel contextual information, while UFZ [49] has made significant progress in aggregating functionally similar objects. Other works optimize class representations from different perspectives: LoG-CAN [37] integrates class context through cascaded refinement and feature fusion, while its successor, LoG-CAN++ [38], introduces an affine transformation to extract local class centers. Despite the significant progress achieved by these methods, their guiding prototypes were derived from the model's own coarse-grained predictions. When faced with high intra-class variance, these coarse prototypes had limited representational capacity and inevitably introduced misleading features [50], [51].

In contrast, our APEM completely abandons the traditional path of relying on model predictions. APEM utilizes ground truth to directly aggregate all diverse representations of a single class from the feature maps, thereby constructing a comprehensive semantic prototype. This approach precludes the introduction of misleading information at its source. To further enhance the adaptability of the class prototypes to complex scenes, we leverage the enhanced features from the SSCM's output to dynamically update these prototypes.

B. State-Space Models

In recent years, the rapid development of SSMs in the field of deep learning has significantly enhanced the capability for long-range dependency modeling and sequential data processing. The introduction of HiPPO initialization has facilitated the integration of SSMs with deep learning, strengthening their ability to capture long-range dependencies [52]. The LSSL model first addressed the challenge of long-term dependence in SSMs [53], but it still has limitations in computational and memory efficiency. Subsequently, the S4 model [54] reduced computational overhead through a novel parameterization strategy, making SSMs more practical. On this basis, SSMs have derived various variants, such as those using complex diagonal structures to enhance temporal modeling [55] or supporting multi-input–multi-output configurations to improve flexibility [56].

In the field of remote sensing, SSMs have attracted significant attention due to their long-range modeling capabilities and linear complexity. A recent research direction has been to extend the powerful capabilities of 1-D sequence processing for application to remote sensing images. RSMamba [34] was the first to apply SSMs to remote sensing dense prediction tasks, validating their feasibility. RSCaMa [35] designed the TT-SSM module by leveraging the potential correlation between Mamba's temporal scanning characteristics and the temporality of the RSICC task, which effectively enhances feature interaction across different timestamps. FusionMamba [36] fused information from different modalities by extending its unidirectional input to be bidirectional. Furthermore, models such as Samba [57] and PyMamba [58] demonstrated the impressive performance of the original Mamba block.

However, most of these works are limited to spatial or modal feature interaction, failing to effectively combine Mamba's advantages in sequential modeling with the guiding role of class semantics. In remote sensing image segmentation, class semantics are core to distinguishing different ground objects, while structural information helps in generating consistent segmentation results. Existing methods often overlook the synergistic effect between these two aspects. To this end, we propose the SSCM module, which adopts a hierarchical cognitive strategy to achieve a synergy between class semantics and structural information, thereby enhancing the model's ability to segment complete class instances.

III. Methodology

In this section, we first provide an overview of the overall architecture of PDSSNet. Subsequently, we will delve into the key components of PDSSNet, including APEM, SSCM, and CSAM, elaborating on their respective roles in addressing the problem of incomplete category segmentation.

A. PDSSNet Architecture

The architecture of PDSSNet is shown in Fig. 2. We adopt a pre-trained ConvNext [59] as the backbone network. ConvNext

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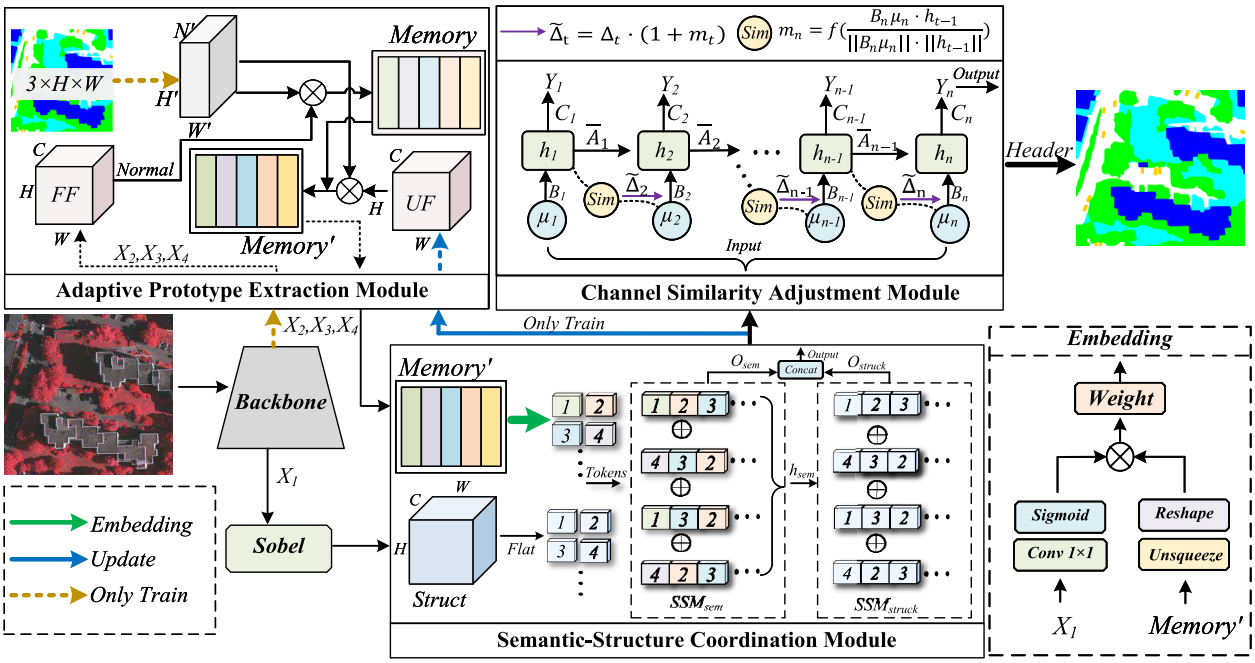


Fig. 2. Architecture of PDSSNet. First, the input image undergoes downsampling, and features from the last three stages are extracted to mix and initialize the APEM. Next, the prototype features output by the APEM and the structural information obtained via the *Sobel* operator are enhanced through feature interaction in the SSCM. The enhanced features are then split into two branches: one branch reversely updates the prototype representation, while the other branch is processed by the CSAM, after which the segmentation head outputs the final segmentation result.

consists of four stages, with each stage performing downsampling. For an input image $x \in \mathbb{R}^{3\times h\times w}$, we first perform four downsampling operations to obtain spatial features at four different scales: $x_1 \in \mathbb{R}^{C\times H\times W}$, $x_2 \in \mathbb{R}^{2C\times(H/2)\times(W/2)}$, $x_3 \in \mathbb{R}^{4C\times(H/4)\times(W/4)}$, and $x_4 \in \mathbb{R}^{8C\times(H/8)\times(W/8)}$. Next, interpolation and convolution operations are applied to fuse x_2, x_3, and x_4

$$FF = \mathrm{Cat}\left(\sigma_{1\times1}\left(\xi\left(x_2\right)\right),\ \sigma_{1\times1}\left(\xi\left(x_3\right)\right),\ \sigma_{1\times1}\left(\xi\left(x_4\right)\right)\right) \quad (1)$$

where $\xi(\cdot)$ denotes the interpolation operation and $\sigma_{k\times k}(\cdot)$ signifies the convolution operation with a kernel of size $k\times k$. The fused features $FF \in \mathbb{R}^{3C\times H\times W}$, FF initializes the storage matrix $Memory \in \mathbb{R}^{N\times C}$ (abbreviation:M), where N is the number of classes and C is the channel dimension of the encoder features (specifically in Section III-B.2). Meanwhile, structural information is generated for x_1 through *Sobel* operations $Struct \in R^{C\times H\times W}$.

Next, before feeding the obtained category semantic and structural information into SSCM, we perform simple spatial position adjustment on the category semantic information to facilitate SSCM's capture of spatial dependencies between categories. The relevant formula is as follows:

$$Em = \mathrm{Reshape}(\mathrm{Unsqueeze}(M)) \quad (2)$$

$$\mathcal{W} = \mathrm{Sigmoid}(\mathrm{Conv}_{1\times1}(x_1)) \quad (3)$$

$$Pos[n\cdot C:(n+1)\cdot C, i, j] = \mathcal{W}[n, i, j]\cdot Em[n] \quad (4)$$

where $E_m \in \mathbb{R}^{N\cdot C\times H\times W}$, $\mathcal{W}[n, i, j]$ represents the confidence weight of class n at position (i, j), and Pos represents the regionalized features of N classes at position (i, j).

Thereafter, category semantic information integrated with spatial positions Pos and structural information $Struck$ is processed by SSCM's hierarchical cognitive strategy to generate enhanced features (specifically described in Section III-B.3). These enhanced features are divided into two branches: one branch reversely updates the memory matrix M to enrich the category prototype representation, and the other branch is fed into the CSAM. Through the designed SimStep mechanism, CSAM increases the update step size of hidden states when encountering similar features, enabling the model to focus on discriminative features. Finally, the output of the CSAM is fed back to the segmentation head to generate high-precision segmentation results.

By integrating the APEM and SSCM, PDSSNet dynamically adapts to complex ground objects in remote sensing images, significantly enhancing the model's category parsing capability. APEM provides accurate category semantic information, while SSCM simulates the human hierarchical cognitive strategy of "semantic understanding first, then structural refinement." In addition, PDSSNet employs CSAM to further improve the model's ability to discriminate confusing features. Through the synergistic effect of these modules, PDSSNet has effectively enhanced the category parsing ability in complex scenes and improved the completeness of category segmentation results.

B. Decoder

In this section, we will introduce the implementation of APEM, SSCM, and CSAM. Before that, we will first introduce the preliminary knowledge of SSM.

1) State-Space Model: The SSM is known as a linear time-invariant system, which maps the input sequence $\mu(t) \in \mathbb{R}$ to

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the response sequence $y(t) \in \mathbb{R}$ through $h(t) \in \mathbb{C}^N$. These mappings are typically represented by

$$h'(t) = \mathrm{A}h(t) + B\mu(t) \tag{5}$$

$$y(t) = \mathrm{Ch}(t) + D\mu(t). \tag{6}$$

Equation (5) is the state transition equation, and (6) is the output equation. $A \in \mathbb{C}^{N\times N}$ is the state transition matrix, $B \in \mathbb{C}^N$ is the input control matrix, $C \in \mathbb{C}^N$ is the output mapping matrix, and $D \in \mathbb{C}^1$ is the skip connection matrix. Given the time step $\Delta \in \mathbb{R}$, the continuous-time model is discretized using the zero-order hold (ZOH) method, resulting in

$$h_t = \bar{A}(h)_{t-1} + \bar{B}\mu_t \tag{7}$$

$$y_t = Ch_t + D\mu_t \tag{8}$$

$$\bar{A} = e^{\Delta A} \tag{9}$$

$$\bar{B} = (\Delta A)^{-1}\left(e^{\Delta A} - I\right)\Delta B. \tag{10}$$

For (10), it can be obtained by using the first-order Taylor expansion

$$\bar{B} \approx (\Delta A)^{-1}(\Delta A)\Delta B = \Delta B. \tag{11}$$

By combining (7), (8), (9), and (11), the output y can be derived and expressed as $y = [y_1, y_2, \ldots, y_t, \ldots, y_L]$.

2) Adaptive Prototype Extraction Module: In remote sensing image segmentation tasks, class information is crucial for feature extraction, especially in scenes with complex ground objects. Existing methods, such as CGGLNet [23], CGGCNet [21], and CG-Swin [24], guide feature extraction by generating coarse-grained class prototypes in their intermediate layers. These coarse-grained prototypes struggle to effectively handle the large intra-class variance. In contrast, our approach utilizes one-hot encoded ground truth, enabling the generation of comprehensive class prototypes without introducing misleading features. The following details the initialization and update processes of the APEM.

a) Initialization: To generate initial representative class prototypes, we perform pixel-level masking on the feature map FF based on the ground truth segmentation map and calculate the average feature vector for each category to form a preliminary class prototype representation. Specifically, for the input $FF \in \mathbb{R}^{3C\times H\times W}$, we obtain $\mathcal{Q} \in \mathbb{R}^{C\times H'\times W'}$ through ℓ_2 regularization and 1×1 convolution. For the ground truth $Y \in \mathbb{R}^{3\times h\times w}$, we perform one-hot encoding and bilinear interpolation to generate $\mathcal{L} \in \mathbb{R}^{N\times H'\times W'}$ with the same scale as $\mathcal{Q}$. To initialize the kth entry $M[k]$ in the storage matrix, we perform average pooling on the masked region by referring to the segmentation mask of the kth class, as shown below

$$M[k] = \frac{\alpha_k \cdot \mathcal{L}[k]\,\mathcal{Q}^T}{\sum \alpha_k \cdot C_k}, \quad \alpha_k = \frac{1}{\log(C_k + 1)} \tag{12}$$

where C_k is the number of pixels belonging to the kth class in the ground truth, $M[k] \in \mathbb{R}^{N\times C}$ is the matrix storing class prototypes, $\mathcal{L}$ is the one-hot encoded ground truth with size $N \times H'W'$, and $\mathcal{Q}$ is reshaped into $C \times H'W'$. Considering the extremely imbalanced class distribution in remote sensing images, as shown in Fig. 3, we introduce a class weight α_k to ensure the contribution of rare classes.

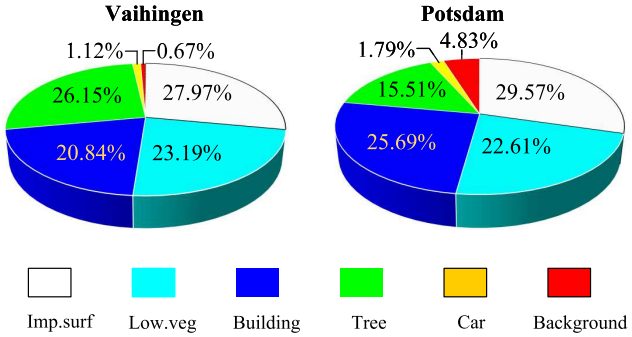


Fig. 3. Datasets' label proportion chart.

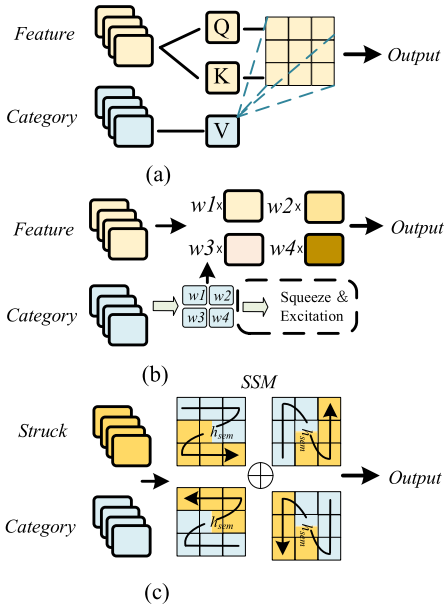


Fig. 4. Three different class-guiding methods are compared. (a) Traditional self-attention method. (b) Generating class weights. (c) Our proposed F-SS2D hierarchical guidance method.

b) Update: To evolve the class prototypes from a static benchmark into a dynamic representation adaptable to complex scenes, we utilize the enhanced features $UF \in \mathbb{R}^{C\times H\times W}$ output by the SSCM to dynamically update them

$$\tilde{\mathcal{A}}[k] = \mathcal{L}[k]\,UF'^T \tag{13}$$

$$M'[k] = \boldsymbol{\beta} \cdot M[k] + (1 - \boldsymbol{\beta}) \cdot \tilde{\mathcal{A}}[k] \tag{14}$$

where $\widetilde{\mathcal{A}}[k] \in \mathbb{R}^{N\times C}$ is the masked feature map, UF' is obtained by performing ℓ_2 regularization on UF and reshaped into $C \times H'W'$, $M'[k]$ is the updated matrix, and the parameter $\boldsymbol{\beta}$ is set to 0.7. It should be noted that the initialization and updating of M are completed during the training phase. Once training is finished, M is considered part of the model's learned parameters and remains fixed during the inference stage. Therefore, during the inference stage, APEM does not perform any updates. Its sole function is to provide its trained class prototypes, M, to the SSCM as a static semantic prior.

3) Semantic-Structure Coordination Module: As illustrated in Fig. 4, most previous category-guided methods rely on self-attention frameworks or generate category weights by

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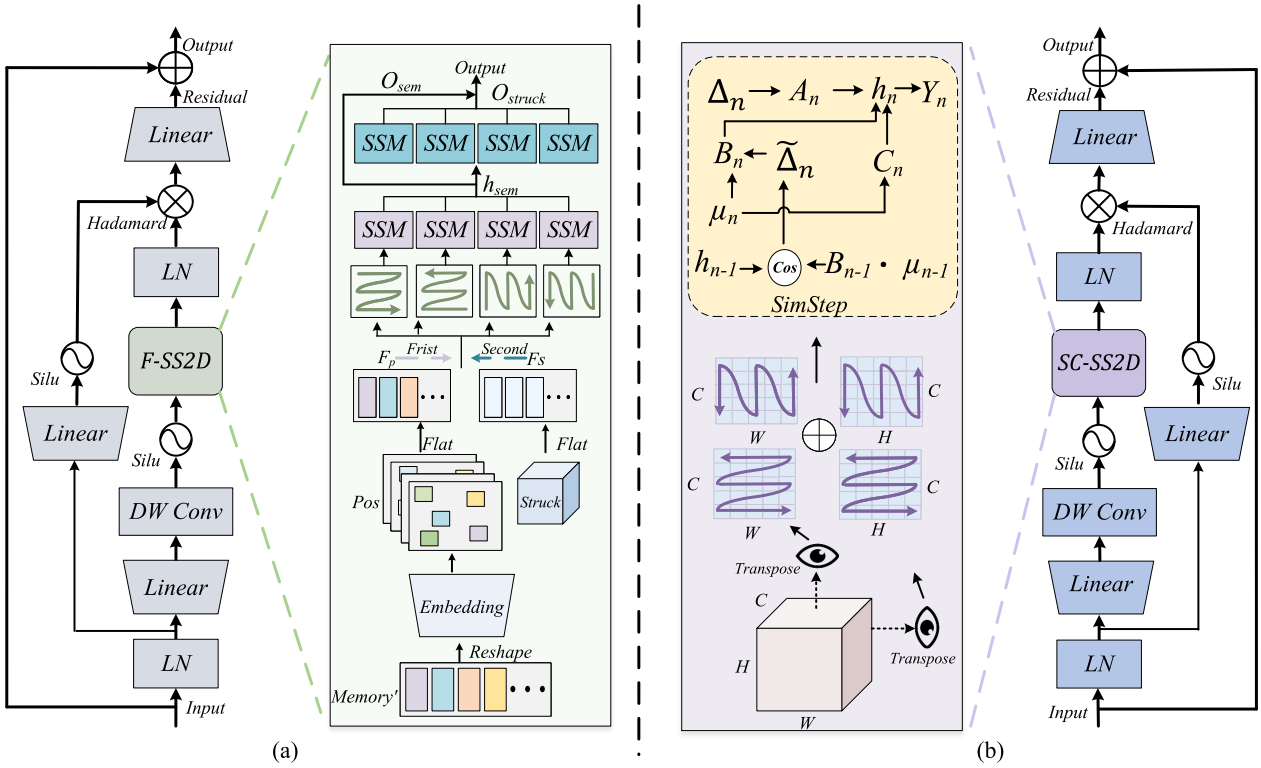


Fig. 5. Structures of SSCM and CSAM. (a) Structure of SSCM takes F-SS2D as the core component. (b) Structure of CSAM takes SC-SS2D as the core component.

compressing category information. The category weight generation approach depends on global spatial statistics, ignoring local details and leading to spatial information loss. Methods based on self-attention frameworks use global *QKV* computation and Softmax to generate weights, but Softmax tends to homogenize weights, reducing discriminability. In high-resolution images, this results in "blurred" feature representations and feature guidance errors [28], [29], [30]. In addition, these methods often overlook class structural information, and this lack of structural constraint leads to the fragmentation of target morphologies. To address this, we designed the SSCM to enable dynamic interaction between semantic and structural information. Inspired by human cognitive laws—where humans first abstract category concepts through long-term memory when perceiving objects, then refine cognition based on observed details, integrate global semantic relationships, and finally iteratively optimize category understanding through feedback [60], [61] SSCM improves the expression of class information by simulating the dynamic cognitive process of humans in complex scenes.

The VSS block in Vmamba can effectively establish dependency relationships in the spatial domain, and we choose to incorporate VSS. As shown in the Fig. 5. The core component of SSCM, F-SS2D, consists of three main parts: the hierarchical cognitive spatial scanning algorithm (HS), SSM establishment, and merging operations. For the input spatially modulated semantic tensor *Pos* and structural information tensor *Struct*, we first serialize them into 1-D sequences, denoted as F_p and F_s, respectively. The semantic sequence F_p is first processed by the SSM SSM_{sem} to establish global contextual dependencies and capture its final hidden state, h_{sem}. This captured state h_{sem} is then used as the initial state for the SSM, SSM_{struct}. This ensures that the processing of the structural sequence F_s is conditioned on the prior knowledge established during the semantic scan. The entire hierarchical process can be expressed by the following set of equations:

$$(O_{\text{sem},k}, h_{\text{sem},k}) = SSM_{\text{sem}}(F_{p,k}) \tag{15}$$

$$O_{\text{struct},k} = SSM_{\text{struct}}\left(F_{s,k}, \text{initial_state} = h_{\text{sem}k}\right) \tag{16}$$

$$Output = \text{Concat}(O_{\text{sem}}, O_{\text{struct}}) \tag{17}$$

where $k \in \{\text{h}, \text{v}, \text{hr}, \text{vr}\}$ represents one of the four scan directions (horizontal, vertical, and their respective reverse directions). O represents the output features.

4) Channel Similarity Adjustment Module: The similarity between category channels is the main cause of category confusion in remote sensing image segmentation [62]. To strengthen the model's capacity to differentiate between similar channels, inspired by RWKV [63] and Hyena [64], a SimStep mechanism based on the SSM is proposed to control the state update of similar channels by dynamically driving the step size through similarity. Specifically, this mechanism dynamically adjusts the step size by measuring the similarity between the hidden state and the current input: increasing the step size allows the model to "skip" or "compress" short-term similar features, tend to update the state quickly, and weaken the cumulative effect of short-term similar inputs, so as to pay more attention to significant

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changes or long-term dependencies in the sequence. It is worth noting that although Mamba has an implicit filtering mechanism, it focuses more on general sequence modeling and long-term dependencies, without explicit processing of similarity [32]. Cosine similarity is employed to quantify the similarity between feature vectors [65].

The structure of CSAM is similar to that of SSCM, but it adopts SC-SS2D instead of the F-SS2D algorithm in the channel dimension. SC-SS2D consists of the spatial scanning algorithm CS [66] and the SimStep mechanism designed by us. As shown in Fig. 5. In the channel modeling approach of the SimStep mechanism, we only process (11) to dynamically adjust the update of historical states according to different inputs. Specifically

$$\widetilde{\Delta}_t = \Delta_t \cdot (1 + m_t) \tag{18}$$

$$m_t = \text{Sigmoid}\left(\text{ReLU}\left(W_s \cdot s_t + b_s\right)\right) \tag{19}$$

$$s_n = \frac{B_n \mu_n \cdot h_{t-1}}{\|B_n \mu_n\| \cdot \|h_{t-1}\|} \tag{20}$$

where W_s and b_s are the weights and biases of the linear layer, respectively. In the selection of nonlinear functions for the SimStep mechanism, we prioritize the ReLU function, whose ability to distinguish between positive and negative features enables the identification of similar features requiring larger step sizes. In terms of temporal complexity, s_t depends on the previous hidden state h_{t-1}. In Mamba's selective scanning, all values of h_{t-1} are precomputed and cached via parallelization techniques. Therefore, the computation of s_t can be integrated into the scanning process without waiting for results from the previous time step, enabling full parallelization [56], [67]. To ensure that our category storage matrix contains more detailed category information, the features after the CSAM do not update the category matrix.

C. Loss Function

For the PDSSNet proposed by us, the entire network is trained using a loss function $\mathcal{L}$, which consists of $\mathcal{L}_{\text{ce}}$ and $\mathcal{L}_{\text{dice}}$. Specifically

$$\mathcal{L}_{\text{ce}} = -\frac{1}{N}\sum_{n=1}^{N}\sum_{k=1}^{K} y_k^{(n)} \log \hat{y}_k^{(n)} \tag{21}$$

$$\mathcal{L}_{\text{dice}} = 1 - \frac{2}{N}\sum_{n=1}^{N}\sum_{k=1}^{K} \frac{\hat{y}_k^{(n)} y_k^{(n)}}{\hat{y}_k^{(n)} + y_k^{(n)}} \tag{22}$$

$$\mathcal{L} = \mathcal{L}_{\text{ce}} + \mathcal{L}_{\text{dice}} \tag{23}$$

where N denotes the number of training samples and K denotes the number of classes. $y_k^{(n)}$ represents the one-hot encoding of the ground truth that sample $n \in [1, \ldots, N]$ belongs to class k, and $\hat{y}_k^{(n)}$ denotes the probability that sample n belongs to class k.

IV. Experiments

A. Experimental Settings

1) Datasets: For our evaluation, we used the Vaihingen, Potsdam, and LoveDA datasets, which are standard benchmarks for remote sensing semantic segmentation. These widely used datasets were selected to ensure the comparability and generalizability of our results, as they cover diverse land categories, such as buildings and road systems, under complex environmental conditions like varied seasons and lighting. Their prevalence in academia facilitates a direct comparison of our work with previous studies.

a) Vaihingen dataset: The dataset comprises 33 high-resolution TOP image tiles, each with an average dimension of 2494×2064 pixels, featuring real orthophoto (TOP), digital surface model (DSM), and normalized DSM (NDSM) data. It includes five foreground land cover categories: impervious surfaces, buildings, low vegetation, trees, and cars, along with a background class: clutter. During the experiment, we carefully chose training samples based on the specific training IDs (ID 1, 3, 5, 7, 11, 13, 15, 17, 21, 23, 26, 28, 30, 32, 34, and 37) provided by the ISPRS competition, using the remaining 17 images for testing. To facilitate processing, these image tiles were cropped into smaller 1024×1024 pixel blocks.

b) Potsdam dataset: The Potsdam dataset offers a more extensive and challenging evaluation scenario, containing 38 high-resolution tiles from Potsdam, Germany. Each 6000×6000 pixel tile features a fine-grained ground sampling distance (GSD) of 5 cm and provides data across four multispectral bands (red, green, blue, and near-infrared), accompanied by a corresponding DSM. According to the official ISPRS protocol, the dataset is divided into a training set composed of 24 specific tiles (IDs: 2_10, 2_11, 2_12, 3_10, 3_11, 3_12, 4_10, 4_11, 4_12, 5_10, 5_11, 5_12, 6_7, 6_8, 6_9, 6_10, 6_11, 6_12, 7_7, 7_8, 7_9, 7_11, and 7_12) are utilized for training, leaving the other 15 tiles for the test set. For processing, these large tiles are partitioned into 1024×1024 patches. Following common practice, the "clutter/background" class is excluded during quantitative evaluation.

c) LoveDA dataset: The dataset is constructed from Google Earth imagery, spanning three Chinese cities: Nanjing, Changzhou, and Wuhan. It comprises 5987 high-resolution optical remote sensing images at 0.3-m ground resolution. Each 1024×1024 pixel image includes seven land cover categories: buildings, roads, water bodies, wasteland, forests, agricultural areas, and background. The dataset contains three spectral bands—red, green, and blue. Unlike Vaihingen and Potsdam dataset processing, LoveDA divides training, validation, and test sets by geographical location to amplify regional diversity. Specifically, the dataset is split into 2522 training images, 1669 validation images, and 1796 official test images. It encompasses both urban and rural landscapes. However, the dataset poses significant research challenges due to multiscale objects, complex backgrounds, and imbalanced class distributions.

2) Implementation Details: All experiments were conducted on a single NVIDIA GeForce RTX 2080Ti (11 GB) GPU, running on an Ubuntu 18.04 system with the PyTorch 1.11 framework. For model optimization, we employed the AdamW optimizer, setting the initial learning rate to $6e^{-4}$ and decaying it throughout training via a cosine scheduling strategy. For the Vaihingen and Potsdam datasets, all images

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TABLE I
RESULTS OF PDSSNET WITH INDIVIDUAL COMPONENTS REMOVED. THE BEST RESULTS ARE INDICATED IN BOLD BLACK

Config	Vaihingen		Potsdam	
	mIoU(%)	F1(%)	mIoU(%)	F1(%)
PDSSNet w/o APEM	83.75	91.04	86.98	92.92
PDSSNet w/o SSCM	83.81	91.08	86.76	92.80
PDSSNet w/o CSAM	84.20	91.32	87.07	92.98
PDSSNet w/o $\mathcal{L}_{ce}$	84.06	91.23	87.02	92.96
PDSSNet w/o $\mathcal{L}_{dice}$	84.36	91.41	87.30	93.11
PDSSNet	**84.68**	**91.60**	**87.55**	**93.25**

were pre-processed by partitioning them into 1024×1024 patches. During the training phase, a suite of data augmentation techniques was applied to enhance model generalization, including random horizontal/vertical flipping, random rotation, and random scaling with factors of 0.5, 0.75, 1.0, 1.25, and 1.5. To ensure a robust evaluation, the testing phase incorporated a multiscale assessment strategy combined with random flip augmentation.

3) Evaluation Metrics: To evaluate the proposed model, we assessed its performance using metrics for segmentation accuracy, including overall accuracy (OA), mean Intersection over Union (mIoU), and $F1$ score ($F1$). We also evaluated its efficiency and complexity using memory access cost (MAC), floating point operations (FLOPs), and inference time. The relevant formulas are as follows:

$$\mathrm{OA} = \frac{\sum_{n=1}^{N} \mathrm{TP}_n}{\sum_{n=1}^{N} \mathrm{TP}_n + \mathrm{FP}_n + \mathrm{TN}_n + \mathrm{FN}_n} \tag{24}$$

$$\mathrm{mIoU} = \frac{1}{N} \frac{\sum_{n=1}^{N} \mathrm{TP}_n}{\sum_{n=1}^{N} \mathrm{TP}_n + \mathrm{FP}_n + \mathrm{FN}_n} \tag{25}$$

$$\mathrm{precision}_n = \frac{\mathrm{TP}_n}{\mathrm{TP}_n + \mathrm{FP}_n}, \quad \mathrm{recall}_n = \frac{\mathrm{TP}_n}{\mathrm{TP}_n + \mathrm{FN}_n} \tag{26}$$

$$F1 = 2 \times \frac{\mathrm{precision}_n \times \mathrm{recall}_n}{\mathrm{precision}_n + \mathrm{recall}_n} \tag{27}$$

where TP_n, FP_n, TN_n, and FN_n represent the true positive count, false positive count, true negative count, and false negative count, respectively, all of which are associated with the specific object of category n (indexed by n).

B. Ablation Experiment

To assess the effectiveness of PDSSNet's individual components, we performed experimental evaluations on the Vaihingen, Potsdam, and LoveDA datasets. The experimental results are averaged over multiple trials and analyzed primarily through two performance metrics: mIoU and $F1$.

1) Components of PDSSNet: To investigate the role of each component in PDSSNet, we conducted a series of experiments. Table I displays the experimental results after module-wise ablation, where APEM represents the adaptive prototype extraction module, SSCM represents the semantic-structure coordination module, and CSAM represents the channel similarity adjustment module, with (w/o) indicating the removal of a component and (+) indicating its inclusion in the experiments. Fig. 6 presents the segmentation performance after deleting a single module of PDSSNet, demonstrating that removing any component negatively impacts network performance. To deeply evaluate the effectiveness of APEM, SSCM, and CSAM, we performed special experiments on the Vaihingen dataset, with the results presented in Table II. These experiments involved using a baseline model (consisting of two convolutional layers connected directly after the ConvNext feature extraction network with a segmentation head) and then individually inserting APEM, SSCM, and CSAM before the segmentation head. Fig. 7 shows the segmentation results after sequentially adding each component to the baseline model, and comprehensive experimental data indicate that each newly proposed component can significantly improve model performance. Table III gives the experimental results of replacing different backbones.

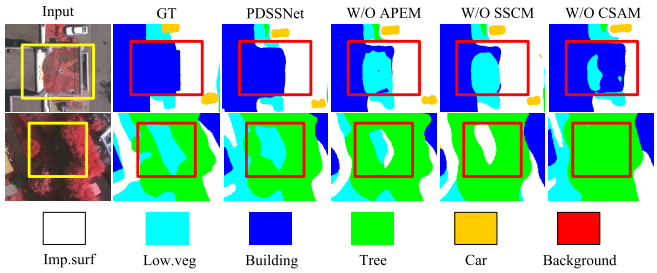


Fig. 6. Partial image of PDSSNet with one module removed on the Vaihingen dataset.

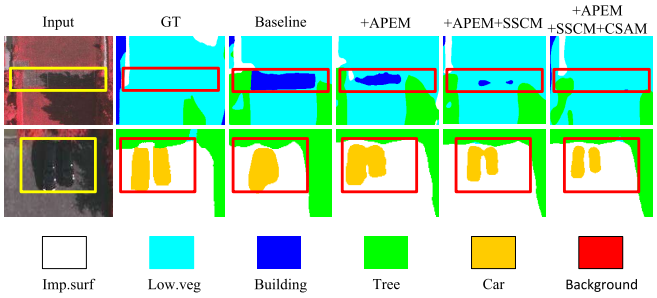


Fig. 7. Baseline adds APEM, SSCM, and CSAM partial enlargement pictures accordingly.

TABLE II
ABLATION STUDY OF PDSSNET WITH DIFFERENT COMPONENT COMBINATIONS

Config	Vaihingen		Potsdam	
	mIoU(%)	F1(%)	mIoU(%)	F1(%)
Baseline	82.60	90.34	85.39	92.00
Baseline + APEM	83.69	91.01	86.53	92.66
Baseline + SSCM	83.73	91.03	86.68	92.75
Baseline + CSAM	83.53	90.91	86.54	92.67

2) Impact of APEM: As shown in the data from Table I, removing the APEM from the PDSSNet model leads to a 0.93% decrease in mIoU and a 0.56% decrease in $F1$ on the Vaihingen dataset. On the Potsdam dataset, the corresponding mIoU decreases by 0.57% and the $F1$ decreases by 0.33%. Conversely, when the APEM is added to the baseline model, the mIoU increases by 1.09% and the $F1$ increases by 0.67%

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TABLE III
COMPARISON OF DIFFERENT BACKBONES

Method	Backbone	Params(M)	mIoU(%)
PDSSNet	ResNet50	25.56	84.17
	ResNext50	25.03	84.32
	ResNest50	25.48	84.29
	Maxvit-Tiny	31.05	84.44
	Swinv2-Tiny	28.33	84.54
	Vim-Small	25.87	84.41
	Vmamba-Tiny	**22.37**	84.52
	ConvNext-Tiny	28.59	**84.68**

TABLE IV
DIFFERENT PARAMETER SETTINGS

β	Vaihingen		Potsdam	
	mIoU(%)	F1(%)	mIoU(%)	F1(%)
0.8	84.48	91.49	87.40	93.16
0.7	84.68	91.60	87.55	93.25
0.6	84.63	91.57	87.20	93.04
0.5	84.65	91.59	87.19	93.04
0.3	84.00	91.20	86.97	92.92
0.1	83.83	91.09	87.08	92.97

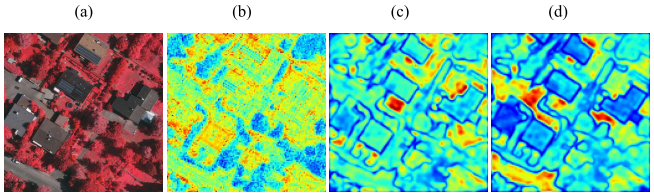


Fig. 8. Feature maps before and after using APEM and SSCM. (a) Remote sensing image. (b) Feature map without using APEM and SSCM. (c) Feature map after applying APEM. (d) Feature map after applying APEM and SSCM.

TABLE V
COMPARISON OF DIFFERENT FEATURE-GUIDED

Config	Vaihingen	
	mIoU(%)	F1(%)
PDSSNet w/o SSCM + CW	84.03	91.21
PDSSNet w/o SSCM + SA	84.24	91.34
PDSSNet w/o F-SS2D + VSS	84.36	91.41
PDSSNet	**84.68**	**91.60**

on the Vaihingen dataset. On the Potsdam dataset, the mIoU increases by 1.14% and the $F1$ increases by 0.66%. The visualization results in Fig. 7 visually demonstrate the effectiveness of APEM in addressing high intra-class variance. After incorporating APEM into the baseline model, the first row shows that for scenarios where the same class exhibits different manifestations, APEM effectively reduces class mispredictions. The second row indicates that for the issue of two cars being mistakenly segmented as one due to shadows, adding APEM achieves clear class separation. This fully demonstrates that by constructing comprehensive class prototypes from GT, APEM significantly enhances both class representation and intra-class feature aggregation capabilities. Further analysis of the visualization results in Fig. 6 shows that when APEM is removed from PDSSNet, the integrity of segmentation results is significantly compromised, with issues such as fragmentation and boundary blurring appearing in target areas. Fig. 8 displays the feature maps before and after the application of APEM. It can be observed that with the use of APEM, the contours of the ground objects become substantially clearer, and the overall color contrast is significantly enhanced. To validate the rationality of the selected update parameter, comparative experiments with different parameter settings were conducted, and the results are shown in Table IV. The experimental results demonstrate that setting $\boldsymbol{\beta}$ to 0.7 yields the optimal value.

3) Impact of SSCM: As indicated by the data in Table I, removing the SSCM from the PDSSNet model causes a 0.87% decrease in mIoU and a 0.52% decrease in $F1$ on the Vaihingen dataset. On the Potsdam dataset, the corresponding mIoU decreases by 0.79% and $F1$ decreases by 0.45%. Conversely, when the SSCM is added to the baseline model, the mIoU increases by 1.13% and $F1$ increases by 0.69% on the Vaihingen dataset. On the Potsdam dataset, the mIoU increases by 1.15% and $F1$ increases by 0.67%. These experimental results collectively demonstrate that the SSCM can improve mIoU by at least 0.76% and $F1$ by at least 0.45%. As the visualization results in Fig. 7 demonstrate, integrating the SSCM further enhances segmentation integrity. This is evidenced by the reduction in erroneous pixel predictions in the first row and the clearer separation of adhered vehicles in the second. This is because the module first utilizes the high-quality prototypes from APEM to establish a global semantic understanding. It then leverages structural information to constrain and refine this initial semantic representation. This refinement process corrects initial misjudgments and ensures the final segmentation result conforms to the true object morphology, thereby significantly improving the completeness of the result.

Fig. 8 visually illustrates the refinement process of feature representations by APEM and SSCM. Initially, in the baseline's original feature map, the activation regions are rather diffuse, and object contours are indistinct. After introducing the high-quality semantic priors from APEM, the basic contours of ground objects (especially buildings) begin to emerge, and the overall contrast of the feature map is enhanced. The addition of SSCM, however, brings a qualitative leap. Not only do the object contours become extremely clear and sharp, but the intra-class feature responses also become more consistent and intense, and the separability from the background is significantly improved.

To validate the effectiveness of the core algorithm F-SS2D compared with category-guided interaction methods, we conducted replacement experiments in PDSSNet: SSCM was substituted with common methods such as self-attention mechanisms and category weighting (as shown in Fig. 4), and compared with the similar method SS2D. The relevant experimental results are shown in Table V (CW denotes the class weight method, SA denotes the self-attention method, and VSS denotes the original VMamba method) and Fig. 9. The visualization results of the latter indicate that after processing with F-SS2D, the color contrast of the features

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TABLE VI
COMPARISON OF SEGMENTATION RESULTS ON THE VAIHINGEN DATASET

Method	Backbone	F1(%)					Evaluation index		
		Imp.Surf	Building	Lowveg	Tree	Car	MeanF1(%)	mIoU(%)	OA(%)
MAResU-Net [68]	ResNet-34	92.91	95.26	84.95	89.94	88.33	90.28	83.30	90.86
LoG-CAN [37]	ResNet-50	93.00	95.67	84.86	90.28	88.27	90.42	82.72	91.20
DC-Swin [17]	Swin-T	93.46	96.00	85.32	90.03	84.88	89.94	82.01	91.29
Mask2Former [46]	ResNet-50	92.86	96.03	84.15	90.50	89.30	90.57	82.99	91.10
MPCNet [47]	ResNet-50	92.76	95.50	84.70	90.40	90.44	90.76	83.27	90.93
LoG-CAN++ [38]	ResNet-50	93.41	95.93	85.26	90.59	89.46	90.93	83.57	91.50
RSMamba [34]	RSMamba-H	93.10	95.61	84.09	90.26	89.42	90.50	82.87	90.98
PyMamba [58]	ResNet-18	92.57	95.41	84.73	90.53	88.27	90.30	82.52	90.91
Samba [57]	-	93.03	95.44	83.84	90.70	89.11	90.42	82.75	90.99
PDSSNet(Ours)	ConvNext-T	**93.46**	**96.28**	**86.38**	**91.39**	**90.50**	**91.60**	**84.68**	**91.94**

TABLE VII
COMPARISON OF SEGMENTATION RESULTS ON THE POTSDAM DATASET

Method	Backbone	F1(%)					Evaluation index		
		Imp.Surf	Building	Lowveg	Tree	Car	MeanF1(%)	mIoU(%)	OA(%)
MAResU-Net [68]	ResNet-34	92.38	95.83	86.65	88.44	96.13	91.89	85.22	90.28
LoG-CAN [37]	ResNet-50	93.36	96.80	87.63	88.83	95.87	92.50	86.26	91.22
DC-Swin [17]	Swin-T	93.26	96.86	87.74	88.68	95.50	92.41	86.10	91.16
Mask2Former [46]	ResNet-50	92.48	96.41	87.53	89.37	96.08	92.37	86.03	90.83
MPCNet [47]	ResNet-50	92.69	96.38	87.30	88.74	96.34	92.29	85.91	90.56
LoG-CAN++ [38]	ResNet-50	**93.82**	97.12	**88.73**	89.60	96.83	93.22	87.50	91.78
RSMamba [34]	RSMamba-H	92.56	96.63	87.43	88.51	95.72	92.17	85.70	90.70
PyMamba [58]	ResNet-18	92.33	96.32	86.99	88.62	96.23	92.10	85.59	90.48
Samba [57]	-	92.87	96.99	87.49	89.16	95.74	92.45	86.17	91.05
PDSSNet(Ours)	ConvNext-T	93.80	**97.30**	88.45	**89.97**	**96.70**	**93.25**	**87.55**	**91.84**

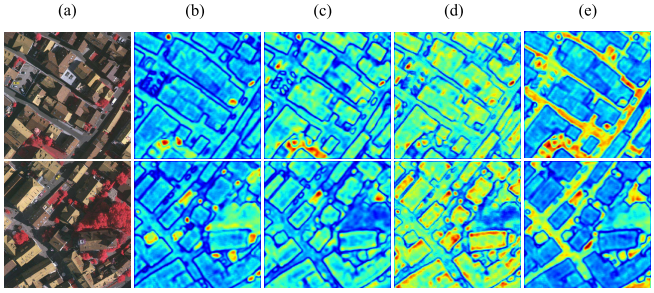


Fig. 9. Feature visualization of three feature interaction methods is provided. (a) Remote sensing image. (b) Category emphasis method. (c) Self-attention mechanism. (d) VSS method. (e) Using the SSCM method.

is significantly enhanced, and there is a greater focus on ground objects that are easily confused or difficult to identify. Specifically, for large regular ground objects, the APEM stage can already outline their general contours through semantic information. For difficult-to-identify ground objects, the first stage performs preliminary positioning based on category semantic information, and the second stage conducts fine-grained adjustment through spatial structure constraints. This result verifies the advantages of the proposed hierarchical cognitive strategy and highlights its advancement in improving the segmentation accuracy of remote sensing images.

4) Impact of CSAM: As shown in the data from Table I, removing the CSAM from the PDSSNet model leads to a 0.48% decrease in mIoU and a 0.28% decrease in $F1$ on the Vaihingen dataset. On the Potsdam dataset, the corresponding mIoU and $F1$ decrease by 0.48% and 0.27%, respectively. Conversely, when the CSAM is added to the baseline model, the mIoU increases by 0.93% and the $F1$ increases by 0.57% on the Vaihingen dataset. On the Potsdam dataset, the mIoU and $F1$ increase by 1.29% and 0.75%, respectively. The experimental results collectively demonstrate that the CSAM can effectively enhance model performance and plays a crucial role in the model's architecture.

As shown in the visualization results of Fig. 6, after removing the CSAM, the model exhibited significant mispredictions when segmenting easily confused categories. As shown in the visualization results of Fig. 7, after incorporating CSAM into the baseline model, the first row demonstrates complete segmentation of category information, and the adhesion between cars in the second row is significantly alleviated. This is because when CSAM encounters similar features, it breaks free from local similarity constraints and infers the category of similar features from other regions, thereby augmenting the model's competence to discriminate confusing features.

Fig. 10 visually illustrates the effectiveness of CSAM. With the introduction of CSAM, the model's response to similar land features, such as tall and low vegetation,

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TABLE VIII
COMPARISON OF SEGMENTATION RESULTS ON THE LOVEDA DATASET

Method	F1(%)							Evaluation index		
	Building	Road	Water	Barren	Forest	Agricultural	Background	MeanF1(%)	mIoU(%)	OA(%)
MAResU-Net [68]	77.75	70.86	77.10	38.59	61.05	68.58	68.34	66.04	50.45	68.92
LoG-CAN [37]	68.74	75.96	69.14	**80.05**	41.08	60.69	68.62	65.94	50.43	69.12
DC-Swin [17]	72.71	72.28	83.30	49.23	57.42	62.17	69.00	66.59	50.83	68.01
Mask2Former [46]	77.51	72.59	83.34	44.07	60.97	68.16	70.93	68.23	52.92	70.82
MPCNet [47]	78.82	71.16	83.45	46.44	55.55	70.38	70.76	68.08	52.80	70.87
LoG-CAN++ [38]	70.99	77.30	71.24	79.79	48.36	58.37	68.05	67.68	52.14	70.22
RSMamba [34]	78.79	72.50	83.21	43.61	**61.52**	69.41	70.88	68.56	53.35	71.16
PyMamba [58]	76.76	66.56	82.28	53.52	54.60	66.81	69.68	67.17	51.40	69.10
Samba [57]	76.01	67.84	81.98	46.37	59.43	66.98	69.61	66.89	51.19	69.19
PDSSNet(Ours)	**78.82**	**73.52**	**85.13**	55.94	60.18	**71.68**	**73.14**	**71.20**	**56.10**	**73.02**

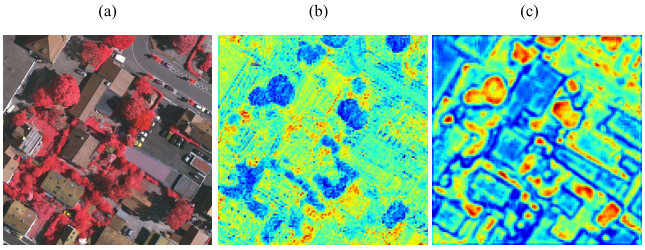


Fig. 10. Feature maps before and after using CSAM. (a) Remote sensing image. (b) Feature map without CSAM. (c) Feature map after applying CSAM.

shifted from being ambiguous and diffuse to sharp and focused. Moreover, the color contrast in the feature maps became more pronounced. This provides strong evidence that CSAM can effectively enhance the model's discriminative ability by increasing the feature distance between similar classes.

5) Impact of the Loss Function: As observed from the experimental results in Table I, different configurations of loss functions have varying degrees of impact on model performance. On the Vaihingen dataset, removing $\mathcal{L}_{ce}$ causes mIoU to decrease by 0.62% and $F1$ by 0.37%, while removing $\mathcal{L}_{dice}$ loss leads to a 0.32% decrease in mIoU and a 0.19% decrease in $F1$. On the Potsdam dataset, removing $\mathcal{L}_{ce}$ results in a 0.53% drop in mIoU and a 0.30% drop in $F1$, whereas removing $\mathcal{L}_{dice}$ has a smaller effect, with mIoU decreasing by 0.25% and $F1$ by 0.14%. Removing $\mathcal{L}_{ce}$, which is used to optimize pixel-level classification and boundary distinction, significantly reduces the mIoU and $F1$ scores. In contrast, removing $\mathcal{L}_{dice}$, which primarily addresses the issues of region overlap and class imbalance, has a lesser impact, indicating that its role is more auxiliary. Overall, combining $\mathcal{L}_{ce}$ and $\mathcal{L}_{dice}$ balances the needs of inter-class discrimination and regional overlap optimization. Removing any of these loss functions will negatively affect model performance to some extent.

6) Impact of Backbone: To reduce the backbone's interference with experimental results, we carried out backbone replacement experiments on the Vaihingen dataset. The experiments selected multiple widely used backbone architectures with similar parameter scales, including convolutional networks (such as ConvNext-Tiny, ResNet 50, ResNext 50, and ResNest 50), Transformer-based networks (such as Maxvit-Tiny [69] and Swinv2 [70] Tiny), and Mamba-based structures (such as Vim-Small [71] and Vmamba-Tiny [33]). The experimental results are shown in Table III, where all backbones exhibit high performance metrics. In terms of the number of parameters, Vmamba-Tiny has the fewest parameters in all experiments, demonstrating a significant computational efficiency advantage. However, when comprehensively balancing the number of parameters and accuracy, ConvNext-Tiny shows the best performance. It can achieve high accuracy while maintaining a low number of parameters. Based on this, we selected it as our backbone.

C. Comparative Experiments

When conducting model comparisons, we selected multiple models as comparison objects, including the multilevel attention residual UNet (MAResU-Net [68]) using linear attention mechanisms, the Swin-Transformer network with a dense connection feature aggregation module (DCFAM) (DC-Swin [17]), the Transformer network Mask2Former [46] using masks for attention calculation, the network with a multiscale prototype-based Transformer decoder (MPCNet [47]), LoG-CAN [37] and LoG-CAN++ [38], which also adopt category-guided methods, RSMamba [34], PyMamba [58], and Samba [57] also use SSM for dense prediction tasks. PDSSNet outperformed the aforementioned models in accuracy on the ISPRS Vaihingen, ISPRS Potsdam, and LoveDA datasets, which are commonly employed for remote sensing segmentation research. Tables VI–VIII present the comparison results between PDSSNet and other models on the Vaihingen dataset, Potsdam dataset, and LoveDA dataset, respectively.

A quantitative analysis of model complexity is presented in Table IX. PDSSNet without CSAM (PDSSNet W/O CSAM) demonstrates excellent computational efficiency. Its inference speed of 71.28 ms is on par with methods like LoG-CAN and significantly outperforms dc-Swin (168.78 ms) and Samba (132.62 ms). Simultaneously, its computational load of 110.92 G FLOPs is not only far lower than that of DC-Swin (277.17 G) and MPCNet (524.95 G) but also superior to that of RSMamba (149.11 G) and Samba (130.39 G). Its MAC

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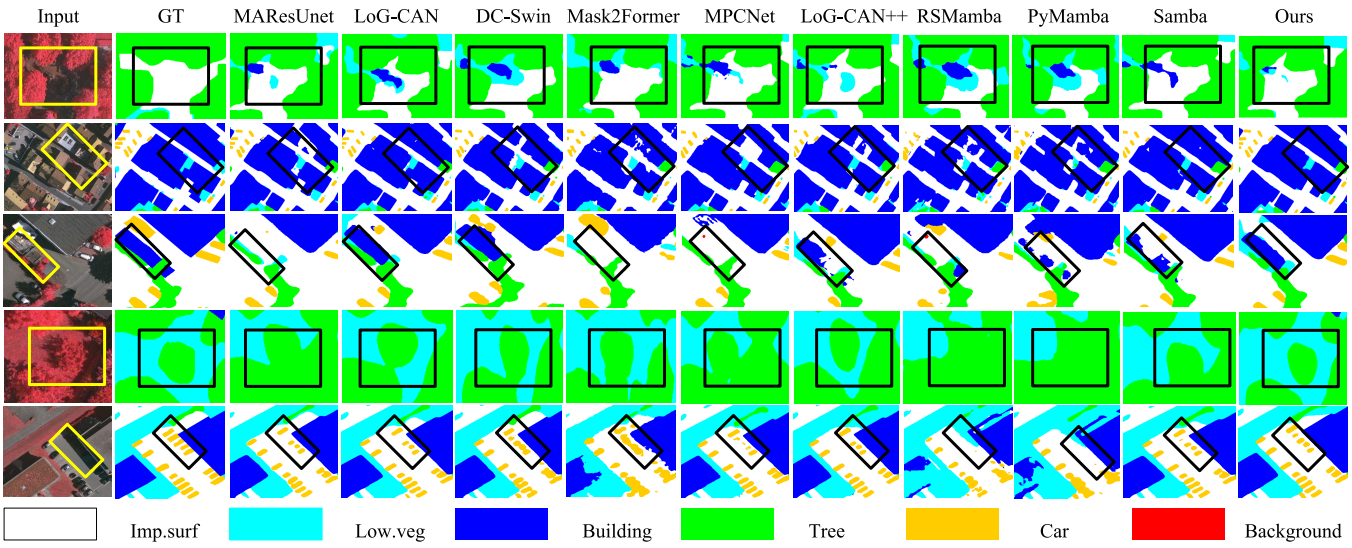


Fig. 11. Local zoomed-in view of the segmentation results on the Vaihingen dataset of PDSSNet and other comparison models.

TABLE IX
COMPARISON OF PERFORMANCE ACROSS DIFFERENT MODELS

Method	FLOPs(G)	MAC(G)	Time(ms)
MAResUnet [68]	140.12	**1.82**	**50.14**
LoG-CAN [37]	99.02	3.45	67.49
Mask2Former [46]	192.62	2.67	263.78
DC-Swin [17]	277.17	8.10	168.78
MPCNet [47]	524.95	3.10	3052.76
LoG-CAN++ [38]	**73.52**	3.16	96.34
RSMamba [34]	149.11	5.00	134.97
Samba [57]	130.39	3.99	132.62
PDSSNet W/O CSAM	110.92	4.12	71.28
PDSSNet(all)	111.31	4.15	230.20

of 4.12 G is also outstanding, significantly outperforming DC-Swin (8.10 G) and RSMamba (5.00 G). However, after integrating the CSAM module, the inference time increases to 230.20 ms. This is primarily because the accelerated functions in the official Mamba library are designed mainly for the spatial scan algorithm and do not yet provide optimized support for the channel scan algorithm employed by CSAM.

1) Results of the Vaihingen Dataset: In the experiments on the Vaihingen dataset, as presented in Table VI, PDSSNet achieved significant improvements in mean $F1$, mIoU, and OA, reaching 91.60%, 84.68%, and 91.94%, respectively. Specifically, PDSSNet outperforms pure convolutional attention-based networks such as MAResU-Net, as well as category-guided method networks like LoG-CAN and LoG-CAN++. Compared with RSMamba, which also uses the SSM, PDSSNet's $F1$ is 1.10% higher, mIoU is 1.81% higher, and OA is 0.96% higher, fully demonstrating the excellent segmentation performance of PDSSNet. The effect is shown in Fig. 11.

This quantitative performance advantage is primarily attributed to PDSSNet's exceptional ability to address the two core challenges of remote sensing imagery. In handling high intra-class variance (as shown in the second row of Fig. 11), PDSSNet maintains the structural integrity of buildings even with variations in lighting and material on rooftops, whereas other methods (like Mask2Former and LoG-CAN) exhibit noticeable internal holes and fragmentation. This directly reflects the synergistic advantage of APEM and SSCM. In dealing with highly similar classes (as shown in the fifth rows of Fig. 11), our model achieves cleaner segmentation when vehicles are parked closely together or confused with shadows, effectively mitigating the adhesion problems found in other methods. This highlights the effectiveness of the CSAM.

2) Results of the Potsdam Dataset: To comprehensively evaluate the segmentation capabilities of PDSSNet, we conducted comparative experiments on the Potsdam dataset. As shown in Table VII, PDSSNet achieved an $F1$ score, mIoU, and OA of 93.25%, 87.55%, and 91.84%, respectively, on the Potsdam test set, with all metrics significantly surpassing those of competing methods. Compared to RSMamba, which also employs an SSM, PDSSNet demonstrated a lead of 1.04%, 1.76%, and 1.06% in $F1$, mIoU, and OA, respectively. Table VII shows the prediction outcomes for the various semantic segmentation methods. The experimental results indicate that the prototype-driven structural constraint method exhibits a significant advantage in the complete segmentation of ground objects.

This quantitative performance advantage is primarily attributed to PDSSNet's outstanding performance in addressing the core challenge of high inter-class similarity, a point fully corroborated by the visualization results in Fig. 12. For example, in the first row of Fig. 12, our model is still able to accurately delineate the clear boundaries of a building even when its rooftop shares highly similar spectral features with the adjacent impervious surface, whereas other comparative

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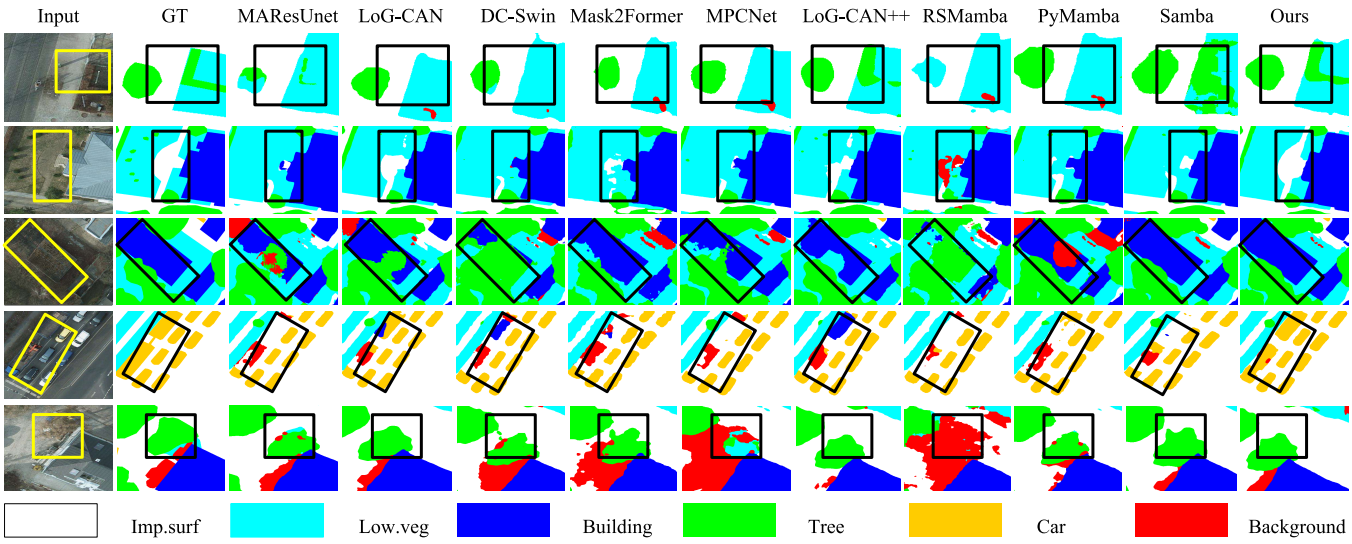


Fig. 12. Local zoomed-in view of the segmentation results on the Potsdam dataset of PDSSNet and other comparison models.

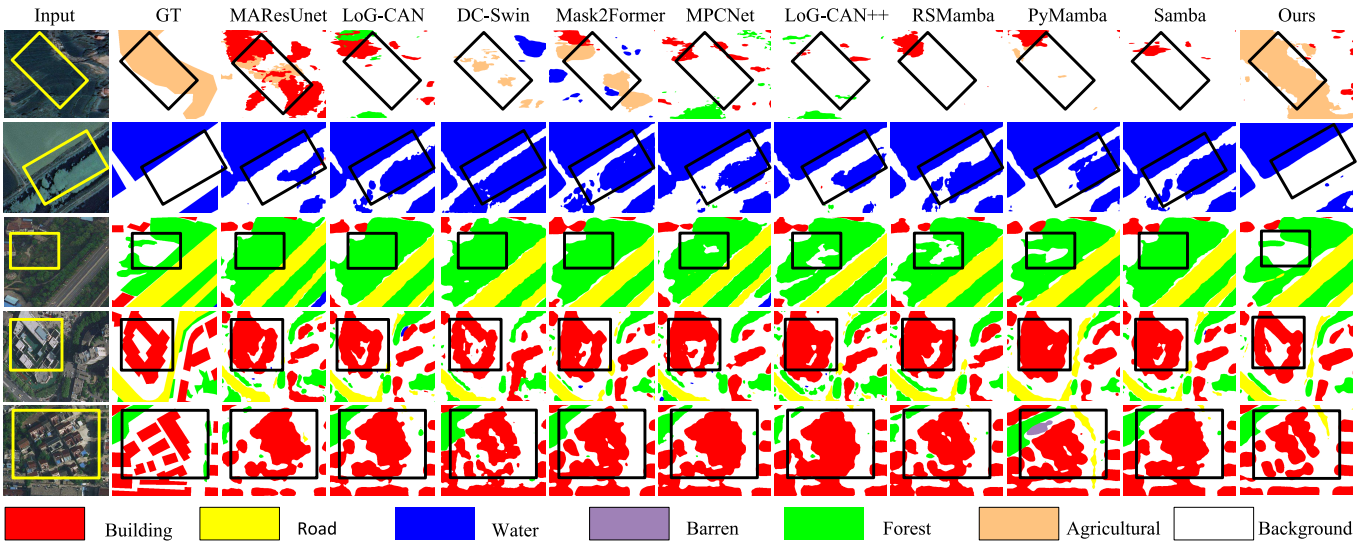


Fig. 13. Local zoomed-in view of the segmentation results on the LoveDA dataset of PDSSNet and other comparison models.

methods produce noticeable class confusion and erroneous segmentation. This powerful discriminative capability stems from the backward update of class prototypes by the structure-refined features, which enables the prototypes to learn more complex, context-aware representations. Furthermore, the Sim-Step mechanism designed within the CSAM module allows the model to "skip" local, easily confusable, similar regions and instead focus on more discriminative boundary information. It is precisely this effective modeling of both high intra-class variance and high inter-class similarity that enables our model to perform exceptionally well on typical ground object classes such as buildings and cars.

3) Results of the LoveDA Dataset: To further validate the generalization capability of PDSSNet, we conducted both quantitative and qualitative analyses of its prediction results on the challenging LoveDA dataset. Compared to Vaihingen and Potsdam, which focus on relatively homogeneous urban environments, LoveDA presents high intra-class variance and high inter-class similarity through its dual urban-rural landscapes spanning different geographical regions. Under these complex conditions, the superiority of PDSSNet is fully demonstrated. From a qualitative perspective, the visual results (as shown in Fig. 13) indicate that, compared to other methods, our model can delineate object boundaries more clearly, effectively reduce class confusion, and significantly improve the completeness and accuracy of the segmentation results. From a quantitative perspective (as shown in Table VIII), PDSSNet significantly outperforms other networks in terms of mean $F1$, mIoU, and OA, achieving scores of 71.20%, 56.10%, and 73.02%, respectively. Its advantage is particularly prominent in the agriculture and water classes. This success is primarily attributed to the semantic reasoning capability of the SSCM. The module's semantic-first strategy enables it to efficiently

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learn the representations of regularly shaped features, such as agricultural areas, once they are identified. Concurrently, it can accurately segment irregularly shaped targets by inferring spatial dependencies between classes, such as the common adjacency of farmland to water bodies. These results fully validate the excellent adaptability and leading performance of PDSSNet in highly heterogeneous and complex remote sensing scenes.

V. Conclusion

To address the problem of incomplete class segmentation in remote sensing images, this article proposed the prototype-driven structure-synergized network. Our proposed APEM established a new method for class modeling. By leveraging ground truth, it constructs comprehensive class prototypes, thereby effectively overcoming the representational limitations of traditional methods when handling high intra-class variance. The SSCM module employs a hierarchical "semantic judgment-structural constraint" cognitive strategy, with the core objective of driving the dynamic alignment of high-level semantics and spatial structure to ensure the morphological integrity of the segmentation results. Finally, the CSAM adopts a novel SimStep mechanism, effectively enhancing the model's ability to distinguish between confusable classes by adaptively adjusting its focus. Experiments have validated the effectiveness of PDSSNet and its individual modules. Its ability to ensure the integrity of segmentation results provides a more reliable data foundation for real-world applications such as land resource planning, urban dynamic monitoring, and environmental protection.

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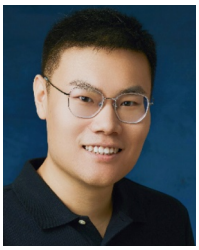
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